COMPASS_TEMPEST_SGW_2023: June

Stephanie J. Wilson

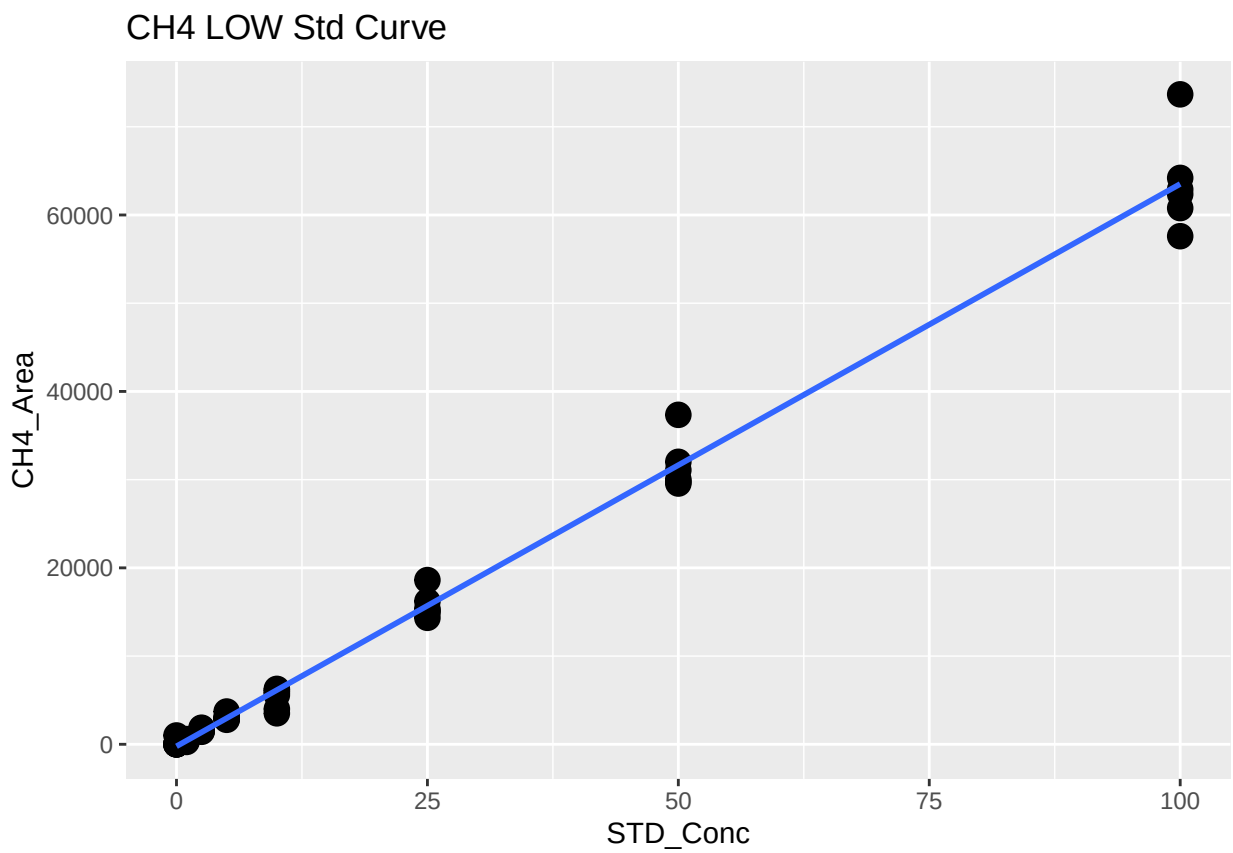
2023-01-14

##Set Up

Read in first data file and assess standard curves

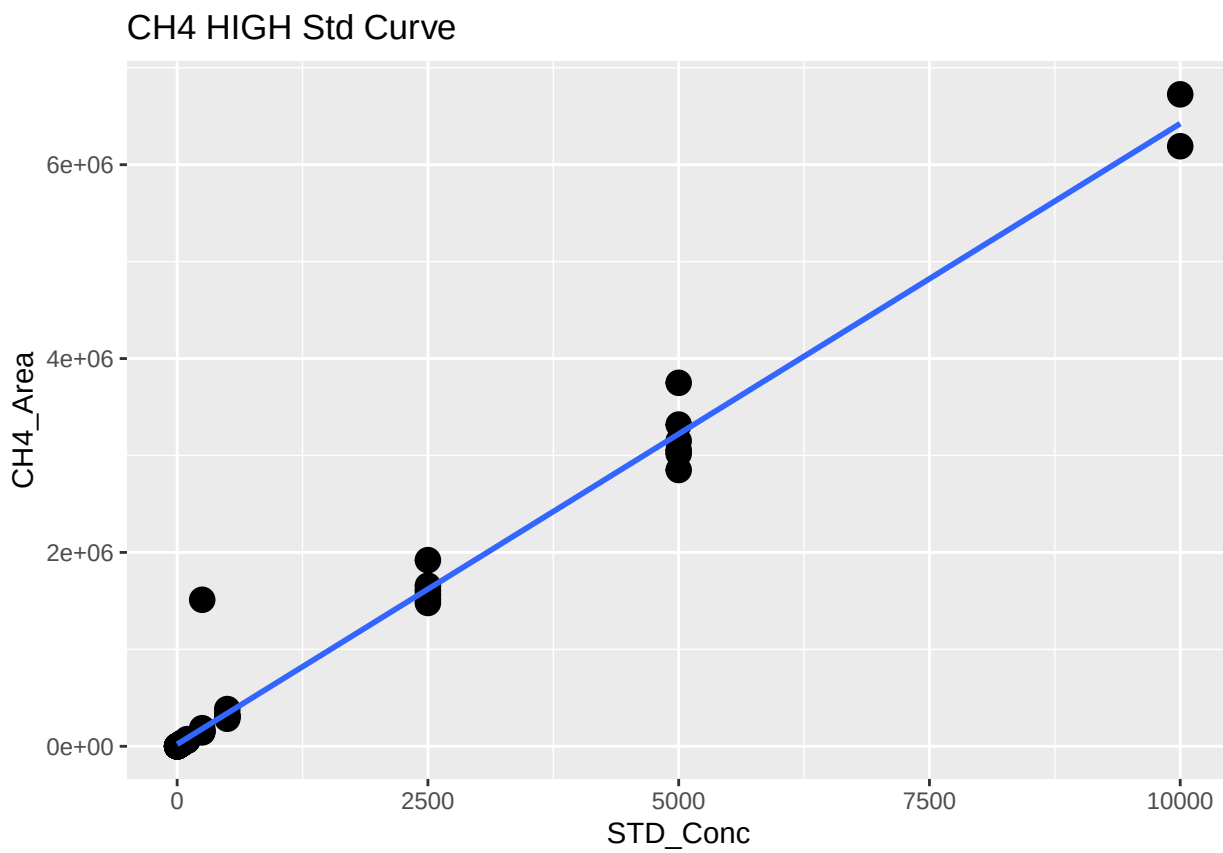
```
##     Machine       User Run_Date Sample_Type Sample1 Sample_Year Sample_Month
## 1 Varian GC Max Wegner 20230627       Blank   Blank        2023         <NA>
## 2 Varian GC Max Wegner 20230627     STD_CH4 STD_CH4        2023         <NA>
## 3 Varian GC Max Wegner 20230627     STD_CO2 STD_CO2        2023         <NA>
## 4 Varian GC Max Wegner 20230627       Blank   Blank        2023         <NA>
## 5 Varian GC Max Wegner 20230627       Blank   Blank        2023         <NA>
## 6 Varian GC Max Wegner 20230627       Blank   Blank        2023         <NA>
##   Sample_Day Event_Stamp Sample_Time Time_Zone G_W Sample_Plot Sample_Grid
## 1         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 2         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 3         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 4         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 5         NA        <NA>          NA      <NA>   G        <NA>        <NA>
## 6         NA        <NA>          NA      <NA>   G        <NA>        <NA>
##   Sample_Depth            Sample_ID Dilution_Factor STD_Conc CO2_Area CH4_Area
## 1         <NA>              Blank_0               1        0     7798        0
## 2         <NA> Blank_0_repeatforCH4               1        0     7798        0
## 3         <NA> Blank_0_repeatforCO2               1        0     7798        0
## 4         <NA>              Blank_1               1        0     3743      603
## 5         <NA>              Blank_2               1        0    21495      398
## 6         <NA>              Blank_3               1        0    27107      368
##   Field.Notes Lab.Notes
## 1
## 2
## 3
## 4
## 5
## 6
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



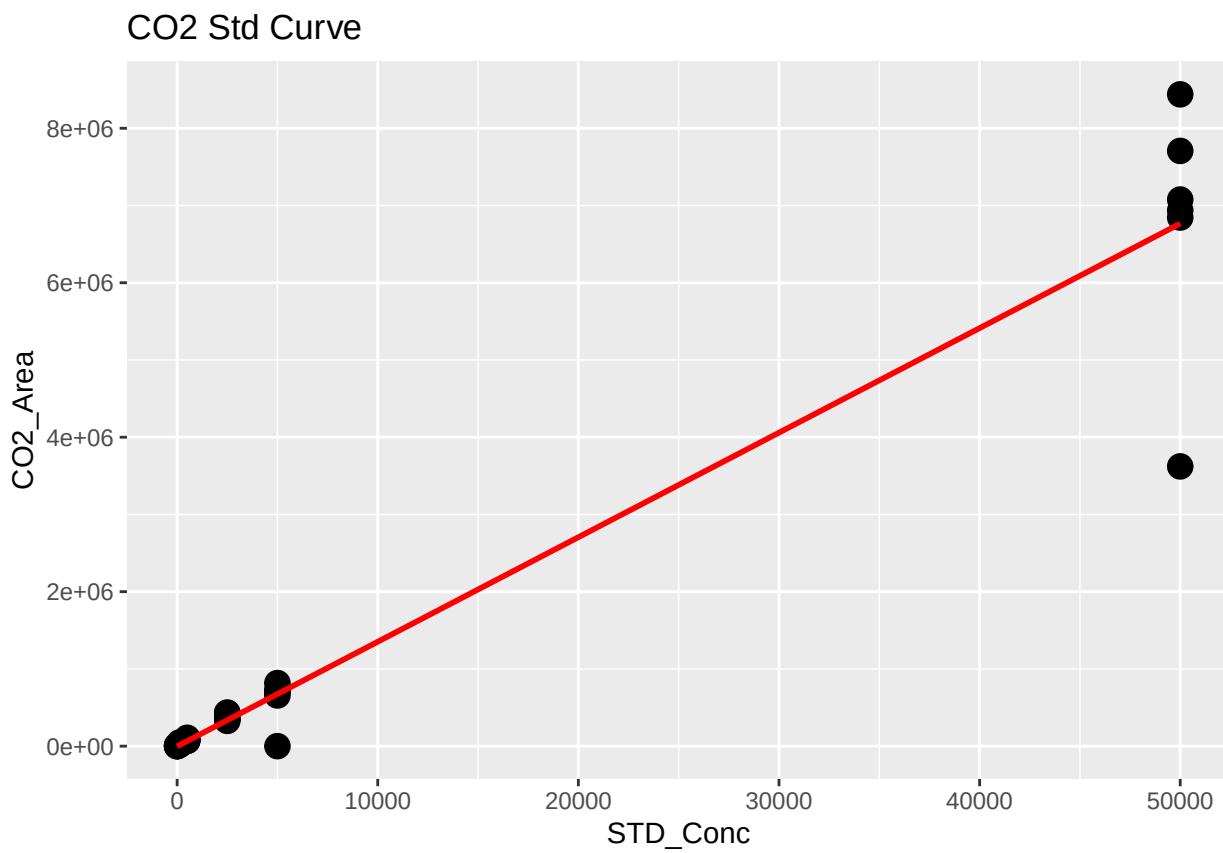

```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5919.8  -568.3    53.9   227.4 10162.2
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -227.390    390.734  -0.582    0.563
## stds_ch4_low$STD_Conc  637.422      9.598  66.409   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2177 on 46 degrees of freedom
## Multiple R-squared:  0.9897, Adjusted R-squared:  0.9895
## F-statistic:  4410 on 1 and 46 DF,  p-value: < 2.2e-16
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

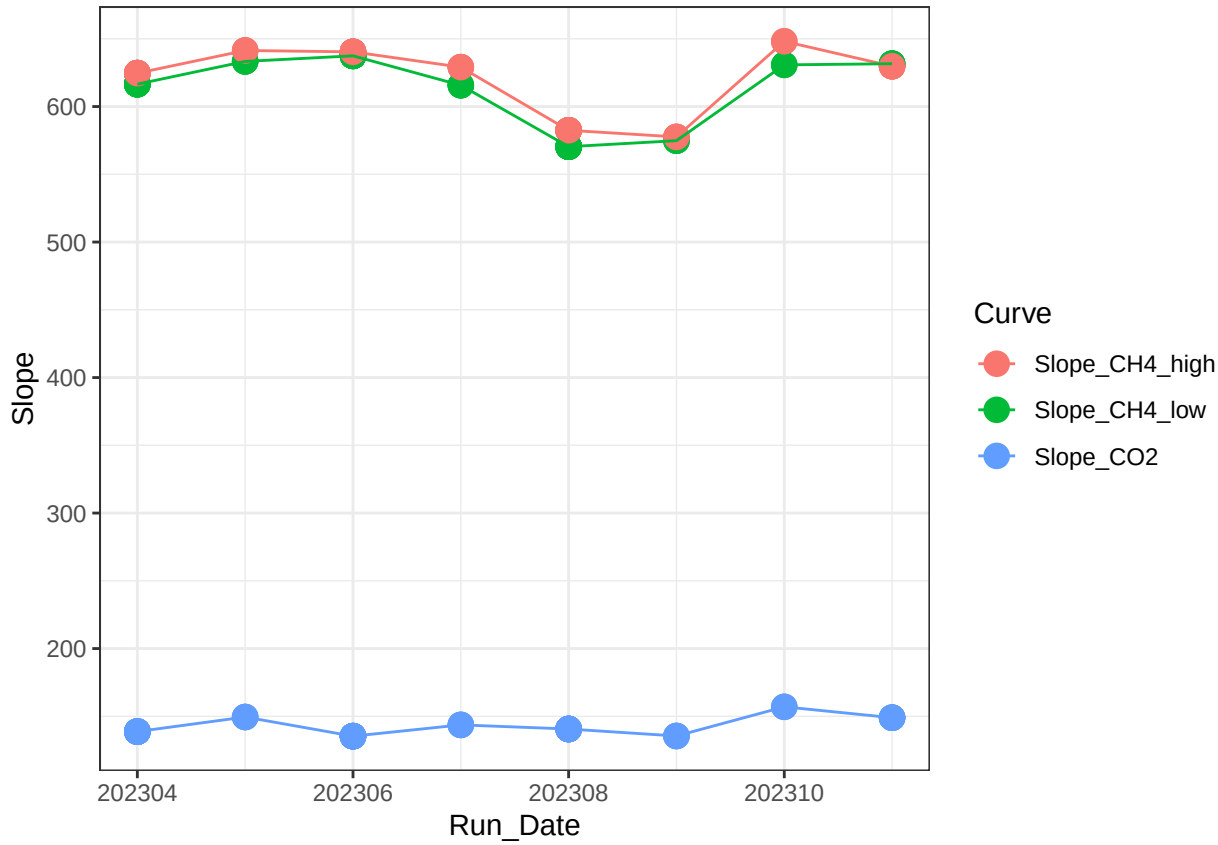



```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -372137  -22837  -20593  -20206 1331099
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        20392.14   24182.99   0.843    0.402
## stds_ch4$STD_Conc    640.36      10.54  60.747   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 189200 on 72 degrees of freedom
## Multiple R-squared:  0.9809, Adjusted R-squared:  0.9806
## F-statistic:  3690 on 1 and 72 DF,  p-value: < 2.2e-16
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3144114     4210     9618    48180  1673233
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -1819.939 126289.672  -0.014    0.989
## stds_co2$STD_Conc    135.359      6.062  22.328   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 656100 on 33 degrees of freedom
## Multiple R-squared:  0.9379, Adjusted R-squared:  0.936
## F-statistic: 498.5 on 1 and 33 DF,  p-value: < 2.2e-16

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 2 2 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 3 3      Slope_CO2 0.9999951 138.7384 5556.86838   202304
## 4 4  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 5 5 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 6 6      Slope_CO2 0.9999951 138.7384 5556.86838   202304
```

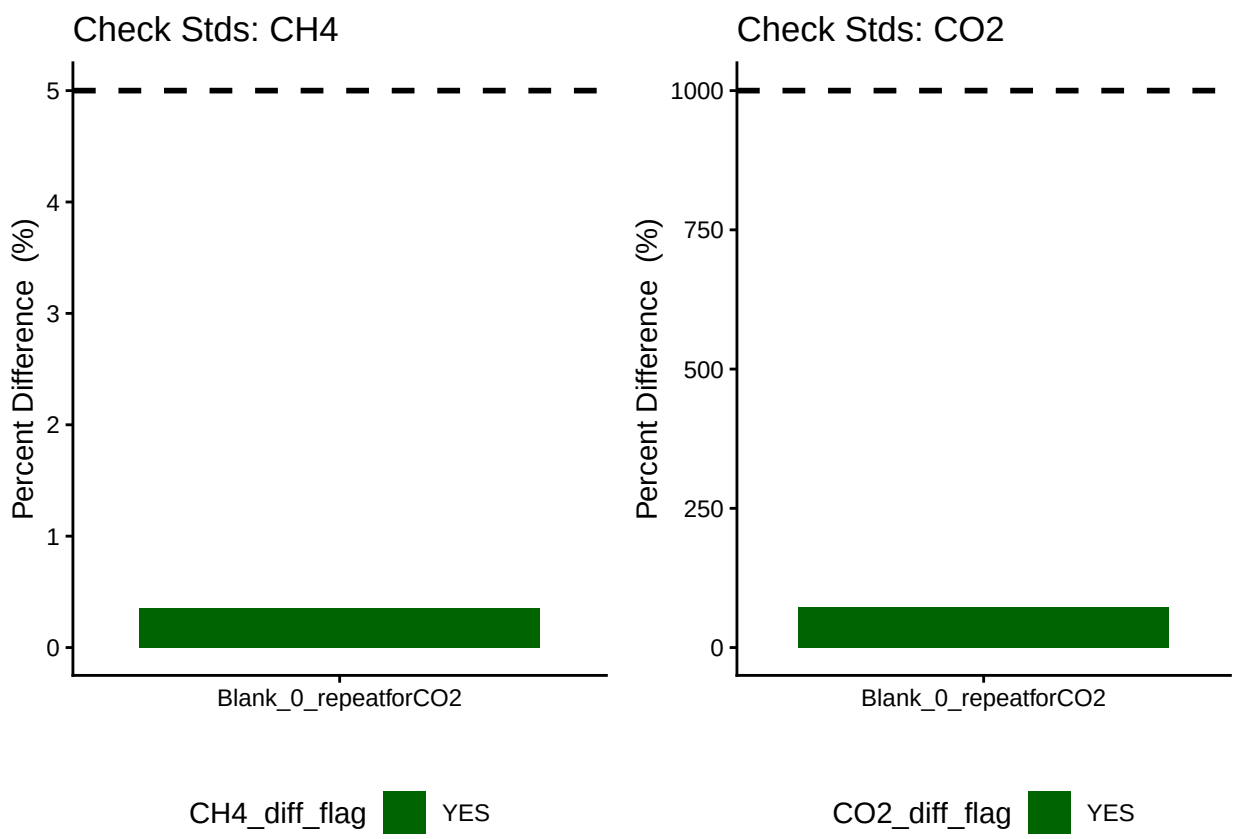


Now calculate the CH4 & CO2 concentrations in ppm

```
#head(raw)

#pull out methane standards
Samples <-  raw %>%
  filter(!str_detect(Sample_Type, "STD_CH4")) %>%
  filter(!str_detect(Sample_Type, "STD_CO2")) %>%
  filter(!str_detect(Sample_Type, "Blank")) %>%
  filter(!str_detect(Sample_Type, "Chk_STD")) %>%
   filter(!str_detect(Sample_Type, "CHKSTD")) %>%
  filter(!str_detect(Sample_Type, "NA"))
#head(Samples)

#Now flag any areas that are above the 100ppm area for CH4
Samples$CH4_Curve <- ifelse(Samples$CH4_Area >71000, "High", "Low")
#head(Samples)

#Calculate CH4 concentrations in ppm
Samples$CH4_Conc_ppm <- ifelse(Samples$CH4_Area >71000, (Samples$CH4_Area-Slope_CH4_high$Intercept)/Slo

#Calculate CO2 concentrations in ppm
Samples$CO2_Conc_ppm <- ((Samples$CO2_Area-Slope_CO2$Intercept)/Slope_CO2$Slope)
```

```
#head(Samples)

##########make flags for any dilutions needed
#highest CH4 standard = 10000
#highest CO2 standard = 50000

Samples$CH4_Flag <- ifelse(Samples$CH4_Conc_ppm >10000, "Needs Dilution", "Within Range")
Samples$CO2_Flag <- ifelse(Samples$CO2_Conc_ppm >50000, "Needs Dilution", "Within Range")
#head(Samples)
```

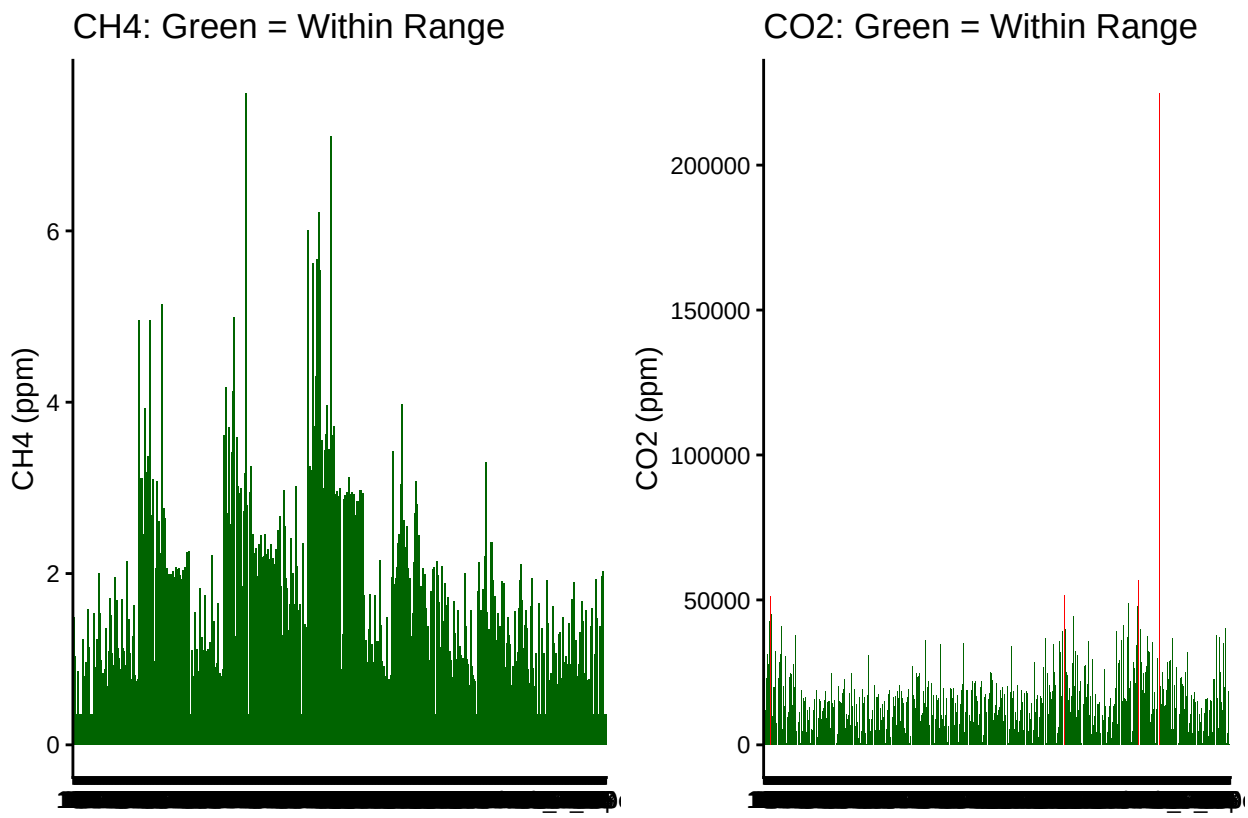
Check the Check Standards



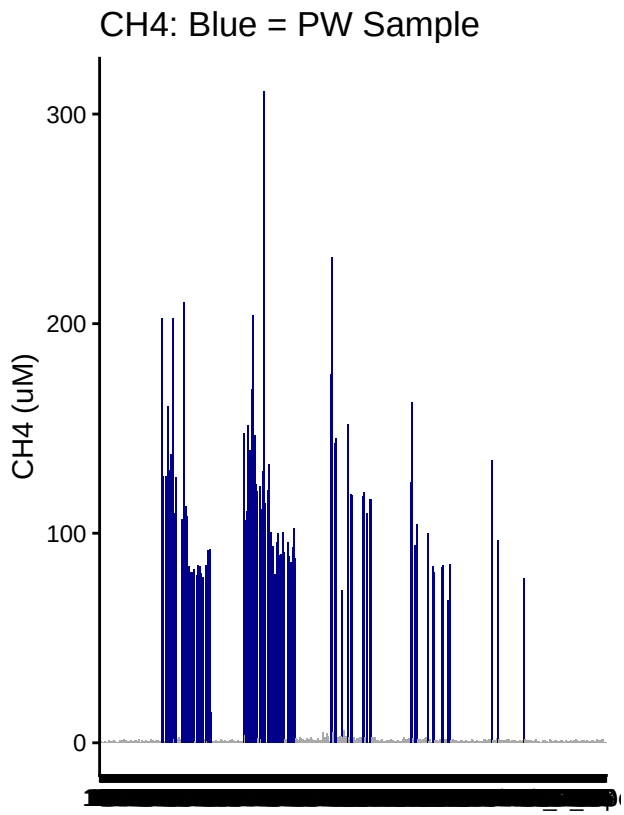
Dilution correct samples

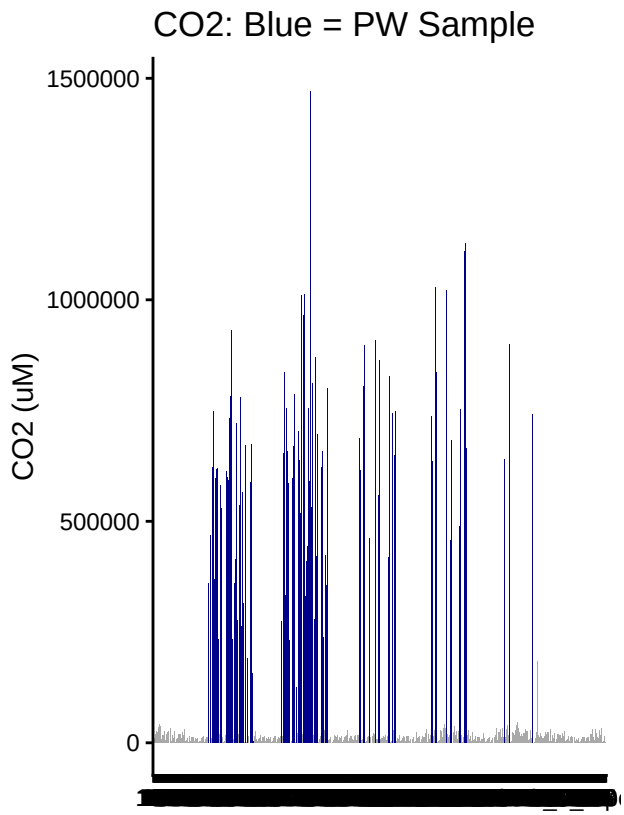
```
#multiply the concentration by the dilution factor
Samples$CH4_Conc_ppm_dilcorr <- (Samples$CH4_Conc_ppm * Samples$Dilution_Factor)

Samples$CO2_Conc_ppm_dilcorr <- (Samples$CO2_Conc_ppm * Samples$Dilution_Factor)

#check results
#head(Samples)
```

```r
#quick first look at the samples
ch4_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CH4_Conc_ppm, fill=CH4_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("darkgreen","red"))+
        theme_classic() + labs(x= " ", y="CH4 (ppm)", title="CH4: Green = Within Range") +
        theme(legend.position="none")


co2_samples <-  ggplot(data = Samples, aes(x = Sample_ID, y = CO2_Conc_ppm, fill=CO2_Flag)) +
       geom_bar(stat = 'identity') +
        scale_fill_manual(values=c("red", "darkgreen"))+
        theme_classic() + labs(x= " ", y="CO2 (ppm)", title="CO2: Green = Within Range") +
        theme(legend.position="none")

ggarrange(ch4_samples, co2_samples, nrow=1, ncol=2)
```



If samples are water calculate gas in water





Write out processed data & slopes

```
#check results
#head(Samples)

#pull out what we need
Samples1 <- Samples[ ,c(1:3,6:17,21, 24:33)]
#head(Samples1)


write.csv(Samples1, "2023 Processed Data/COMPASS_TEMPEST_SGW_202306_Processed.csv")
```

#end